中南大学 邓迪国际学院
DUNDEE INTERNATIONAL INSTITUTE OF CENTRAL SOUTH UNIVERSITY

DI31013 – Engineering Design Project

# Fire Extinguisher Trolley

Group: 05

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| Electrical Circuit and PCB Design | Software development Microcontroller implementation | Mechanical Design Experiments and Results | Future work Abstract Model structural | Mechanical Design: Component design |

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| | | Lessons Learned | testing System analysis | Introduction Conclusion |

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**Abstract**

This report outlines the development of a Bluetooth-controlled firefighting car. The project started with structural modelling using SolidWorks, followed by mechanical reliability analysis conducted in ANSYS Workbench. The next stage is designing and fabricating the PCB, after which key components, including a fan, flame sensor, and water pump, were assembled onto the car base. Programming was then carried out using a mbed microcontroller, and extensive debugging and testing were performed to ensure functionality. The final result is an intelligent car that can be controlled via a mobile phone and is capable of extinguishing fires using both a fan and a water pump.

## Introduction

With the swift advancement of the economy, incidents of fire have become increasingly common in various hazardous environments. Investigations into historical catastrophic fire events reveal that there has been a significant lack of intelligent mobile firefighting equipment. In many cases, traditional tools such as standard fire extinguishers and firefighting vehicles have proven inadequate, often unable to extinguish flames swiftly enough to prevent extensive damage or loss of life. In response to these challenges, intelligent firefighting robots have emerged as a crucial innovation in the realm of fire safety and rescue operations. These advanced robotic systems are designed to navigate complex environments, identify fire sources, and deploy effective firefighting measures autonomously or with minimal human intervention. Their ability to access hard-to-reach areas and operate in hazardous conditions makes them invaluable assets during emergencies (Wu et al., 2021).

In traditional intelligent firefighting robots, the suppression systems integrated are quite sophisticated and versatile. They feature water-based fire monitors that can effectively direct streams of water onto flames, as well as specialized foam nozzles that create a thick blanket of foam to smother fires and prevent re-ignition. To enhance mobility and reach, the robots are equipped with nozzles mounted on articulating arms, allowing for a wider range of motion and greater precision in targeting hotspots. Additionally, the inclusion of a water fog system helps to reduce heat and smoke, improving visibility and safety for both the robot and nearby personnel. These advanced suppression systems are not only operated remotely but also utilize a wireless connection to relay critical data from various onboard sensors back to the operator. This real-time communication aids navigation and enhances fire suppression efforts (Lattimer, 2015).

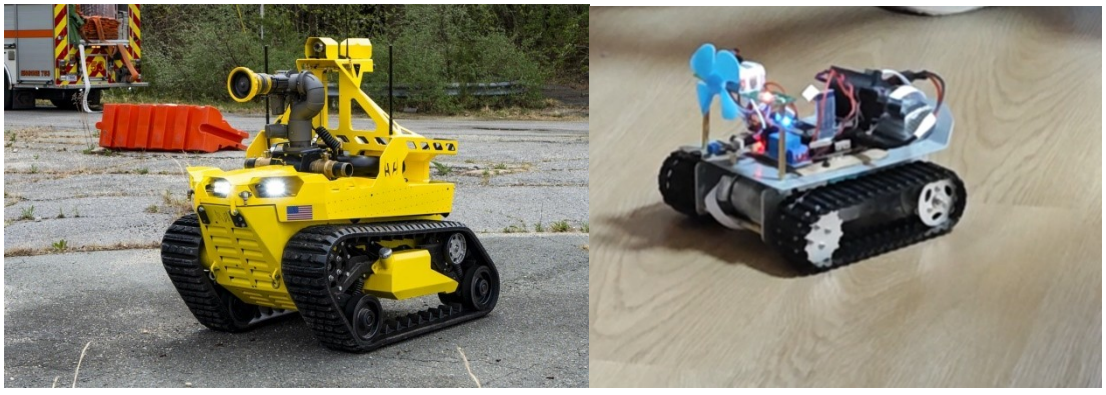


Figure 1 - Firefighting robots images. [1] a) Traditional intelligent firefighting robots. b) Our group's design.

However, traditional intelligent firefighting robots are usually very large and costly because of their sophistication. Based on these characteristics, our team designed a more flexible and cost-effective fire extinguishing car. The car retains the basic functions of traditional robots and improves on their shortcomings. It is notably more compact and flexible compared to other firefighting robots. This compactness allows it to maneuver through challenging environments that may be inaccessible to larger machines. One of the car's significant advantages lies in its ability to operate in tight spaces or through narrow openings, which is crucial in situations where traditional firefighting equipment may struggle to reach. Moreover, the car can be controlled through a mobile phone, which makes rescue more convenient. One of the stands-out features of the car is its ability to enhance productivity, safety, efficiency, and overall quality when performing firefighting tasks. By reducing operations and providing real-time support, the car enables firefighting teams to respond more effectively to emergencies.

The user engages with the mobile application to transmit electromagnetic signals, which are subsequently captured by the Bluetooth module integrated within the designated vehicle. This information is relayed to the mbed platform via the circuit and PCB board. When the device is started, the electronic components on the circuit board begin to work together. The processor is responsible for processing various data, the memory is used for storing and reading information, and the input-output interface is responsible for interacting with the outside world. The mbed processes the incoming data and executes specific commands as dictated by the programmed logic, ultimately facilitating motor control and extinguishing the fire.

## Mechanical Design

### 1. Component design

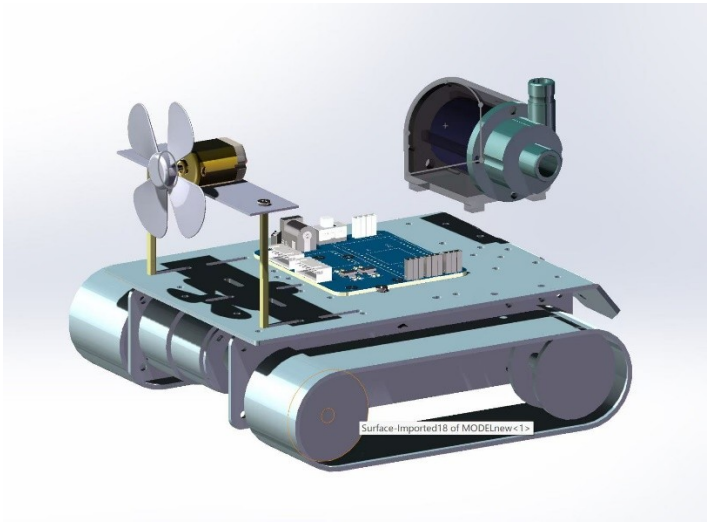



Figure 2 - Car

The car design encompasses several key components: water pump, fans, PCB boards, the chassis, wheels, and tracks. The PCB is intricately populated with electronic elements such as resistors, capacitors, and integrated circuits, collectively constituting the heart of the electronic systems. Through sophisticated circuit interconnections, these components facilitate precise current regulation and the swift transmission of signals. The circuit board is firmly affixed to a base equipped with rollers, enhancing the device's stability while enabling agile movement. Positioned atop the circuit board, the fan plays a pivotal role by generating the essential airflow required for effective fire suppression.

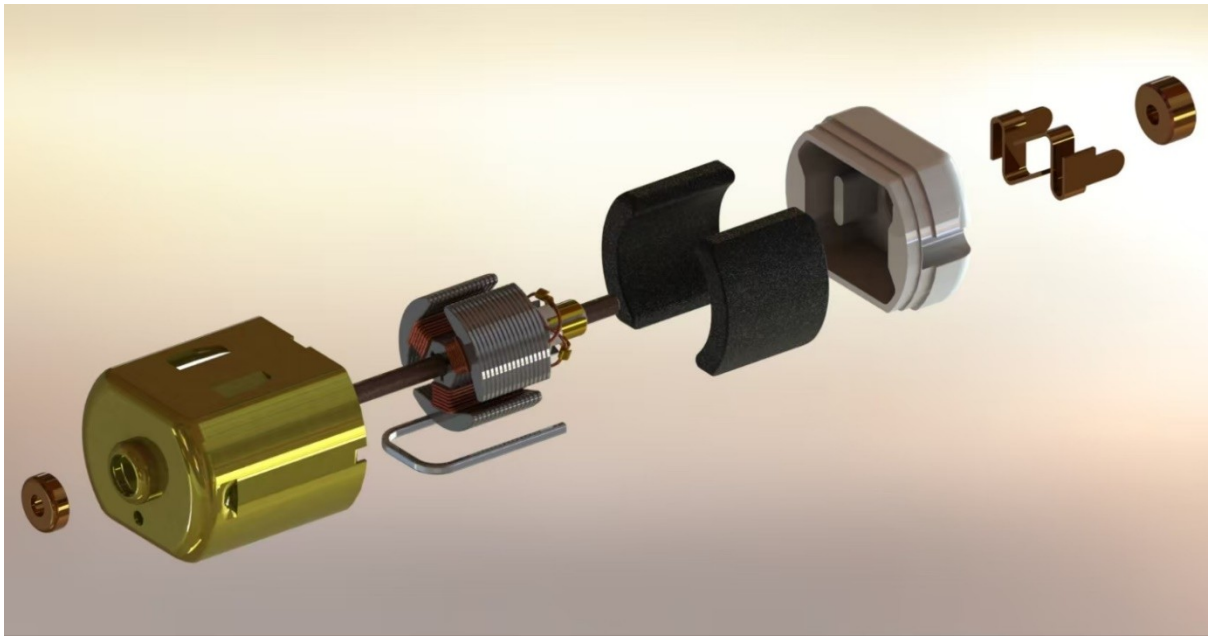


Figure 3 - Electric motor of fan

This is a disassembled diagram of the electric motor, detailing its key internal components and their functions. The diagram mainly includes the following main components.

Motor shaft: As the core component of an electric motor, the motor shaft carries the motor coil and rotor, ensuring that they can rotate smoothly. Addly, the shaft plays a vital role in transforming internal rotational energy into usable mechanical power.

Motor casing: The motor casing serves to protect and secure the internal components of the motor, providing a stable and sturdy outer shell to prevent external interference and damage.

Motor bearings: Located between the motor shaft and the motor housing, the motor bearings provide support and effectively reduce friction, ensuring that the motor shaft remains stable and has low losses during rotation. To minimize wear and prolong the motor's lifespan, high-quality bearings are essential.

Motor coil: As the energy converter of the electric motor, the motor coil is responsible for converting the input electrical energy into mechanical energy. This conversion process drives the motor rotor to rotate, thereby generating power.

Motor stator: The motor stator is fixed on the motor housing and usually includes motor coils, magnets and other fixed components. Its main purpose is to produce a rotating

magnetic field when electric current passes through the coils. This magnetic field interacts with the rotor, leading to its rotation and the generation of mechanical motion.

Motor rotor: The motor rotor is connected to the motor shaft, and through the rotation of the motor shaft, the motor rotor can drive external loads to achieve the output function of the motor. As the rotor engages with the magnetic field generated by the stator, it transforms electrical energy into mechanical energy.

The collaborative work of these key components ensures the efficient and stable operation of the electric motor, providing a continuous source of power for various mechanical equipment (FUKUTA,2024).

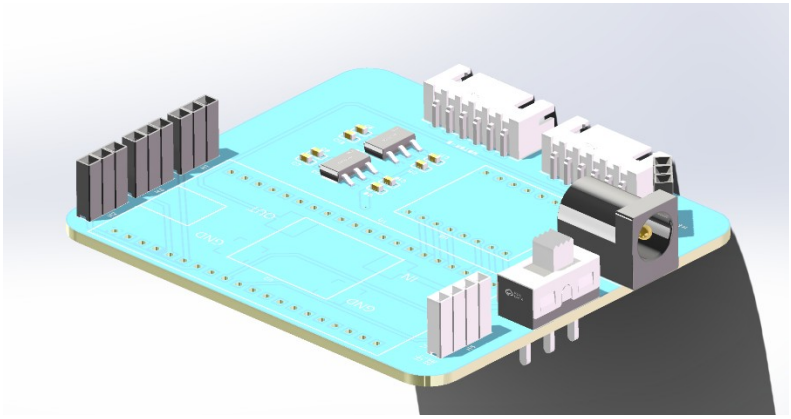


Figure 4 - PCB board

This is the PCB board of our group car, which includes the following main components.

Electronic components: Multiple rectangular gray and white components that are integrated circuits (ICs) or other critical electronic components that are the core of a circuit board and responsible for performing various electronic functions.

Connectors: Multiple black and white circular connectors used to connect components on circuit boards, ensuring smooth transmission of current and signals.

Grounding terminal: The part marked with the word "GND" represents the grounding terminal, which is an important component of the circuit board used to ensure the stability and safety of the circuit.

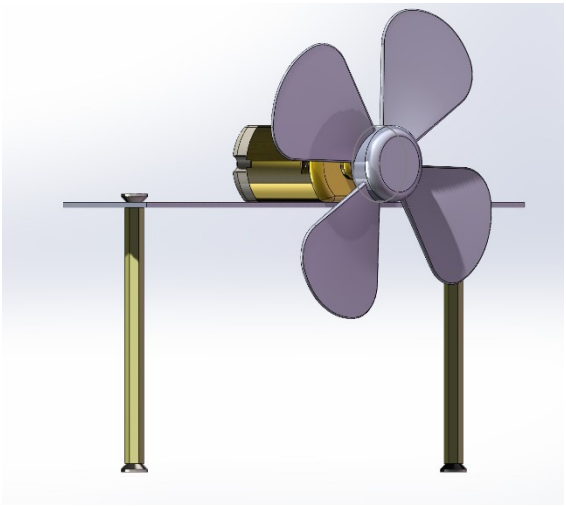


Figure 5 - Fan and electric motor

This is a detailed picture of the fan, with the electric motor behind it. Electric motor drives the fan to rotate.

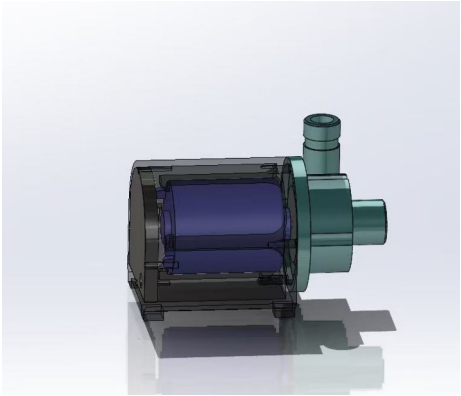


Figure 6 - Water pump

The water pump in this picture contains the following key components:

Main structure: Composed of black and blue plastic or metal materials, it forms a sturdy frame for the water pump.

Cylindrical core component: located at the centre of the main body, responsible for conveying and compressing water flow, and is the key to the operation of the water pump.

Green pipeline: Connected to the top of the main body as an inlet or outlet, ensuring smooth introduction or discharge of water flow.

Black protective cover: located at the top of the main body, used to protect internal components and facilitate maintenance.

A small water pump primarily consists of several key components: the motor, pump body, impeller, seals, and water outlet. The motor serves as the power source, converting electrical energy from the power supply to drive the impeller's rotation. The impeller is the essential part of the pump, creating centrifugal force that draws water into the pump body. Seals are critical for preventing water leakage, while the water outlet allows the discharged water to exit the pump.

The operational principle of a small water pump relies on centrifugal force. Once the motor is activated, the impeller spins to create this force, which pulls water into the pump body before expelling it through the outlet. Throughout this process, seals are vital in preventing leaks, ensuring the pump functions correctly （SHENPENG, 2023）.

## 2. SOLIDWORKS Model

Using SolidWorks (SW) to implement the design model, the following is the SW software section.

The trolley has the following components: the trolley body, the PCB board, the flame sensor, the fan module and the water pump module. Since the flame sensor cannot be exported, it is not reflected in the SW.

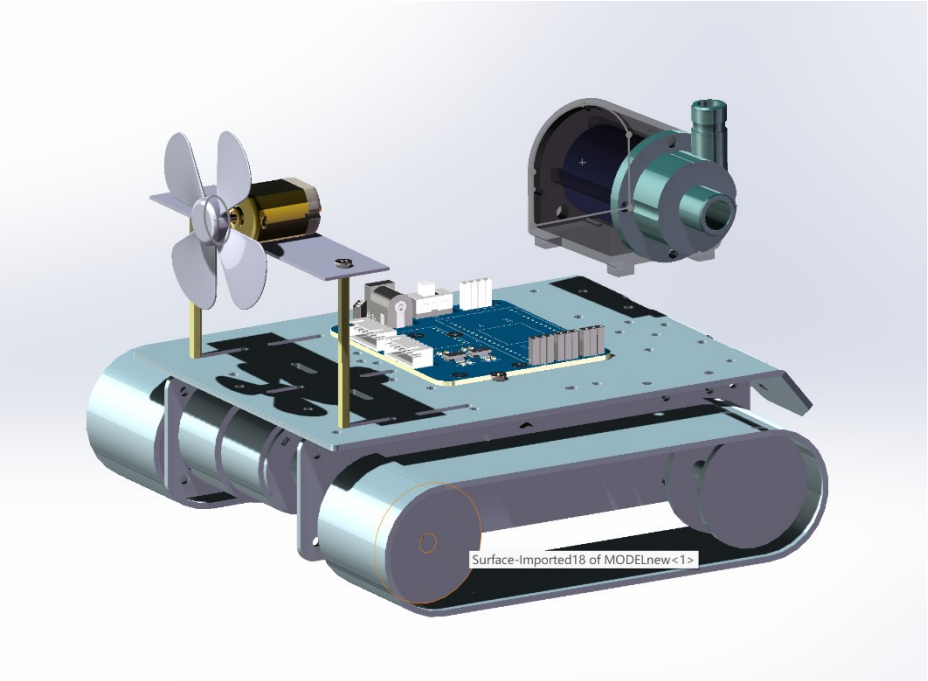



Figure 7 - trolley

The first is the car model part, which mainly has the car bottom plate, two motors, four wheels and two tracks. The motor is placed on the left front wheel and the right rear wheel of the trolley, which is the driving wheel. Their profile is not actually a complete arc, but rather a gear-like engagement with the track, which pushes the track to turn. The holes in the bottom plate of the trolley are mainly used to fix the screws and nuts. Because the wheels are omitted in the modeling, the tracks cannot follow the rotation of the wheels, so the motors added at both ends of the wheels cannot be simulated only by the basic structure.

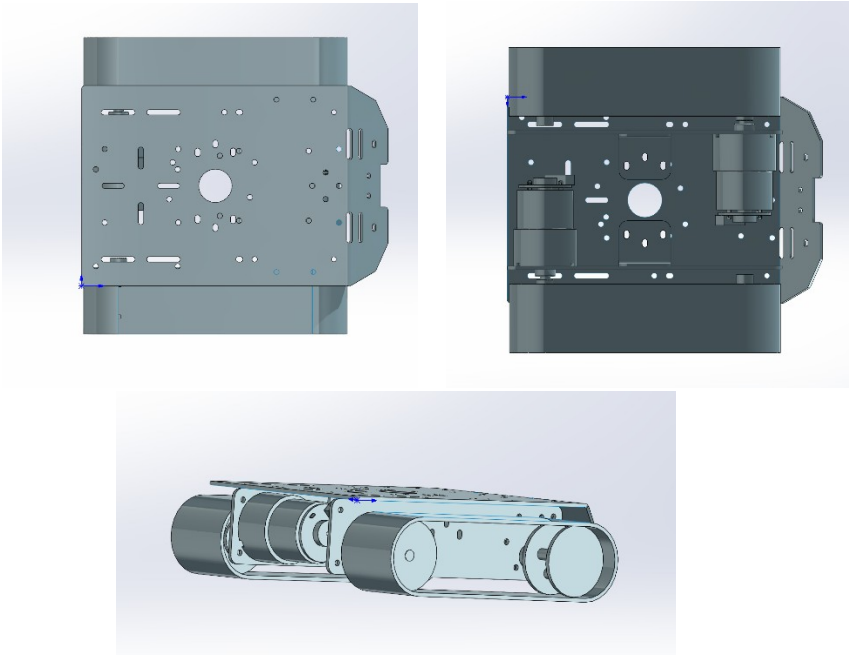


Figure 8 - a) Board. b) Motor. c) Car.

The second part is fan module, mainly consisting of the DC generator, the fan and the base plate where the fan is mounted. The DC generator is a motor that drives a fan. It mainly consists of bushing, contact, housing, magnet, spacer, cover and rotor. The fan consisting of four blades, which is assembled with the DC generator. The fan is a separately drawn part, not an assembly. Also, the notch on the back of the fan needs to be in concentric circles with the rotor part of the motor and locked. This allows the fan to follow the motor.

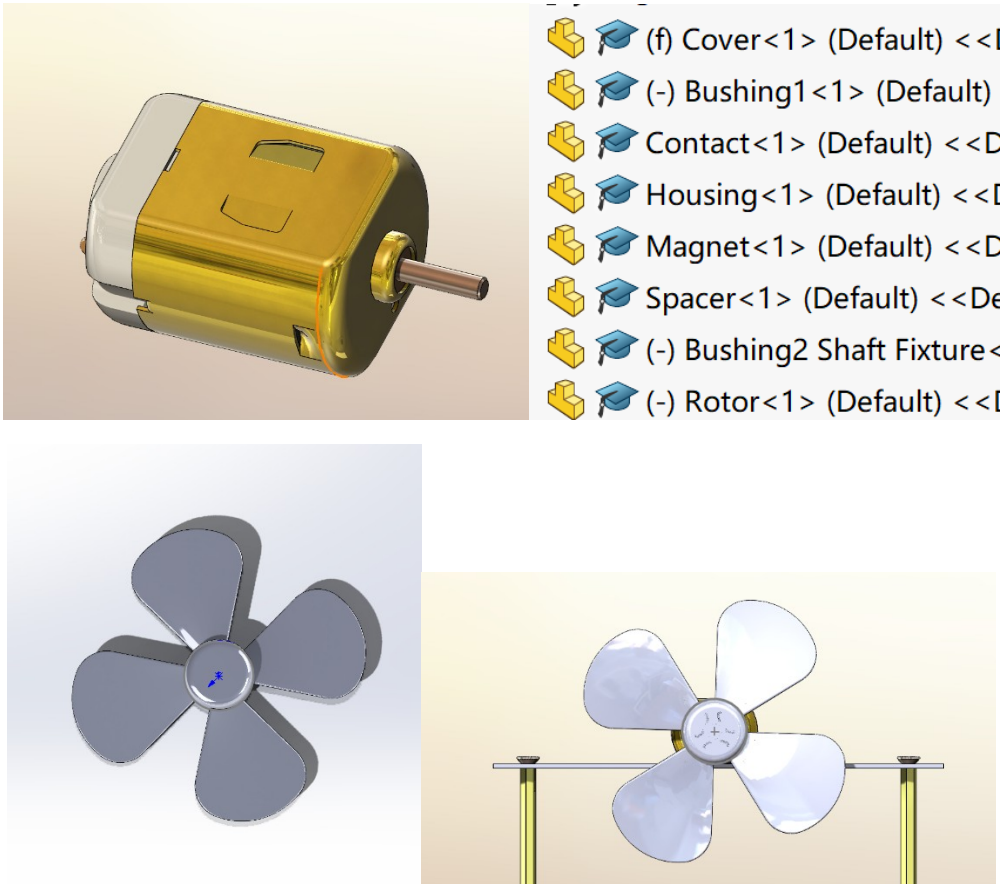



Figure 9 - Fan and Motor

The third part is the standard parts, and the main ones are hex nuts with an inner diameter of 3 mm and screws with the same diameter as the nuts and a length of 5 mm. The standard parts are assembled on the floor of the trolley to secure the trolley. There is also a hexagonal copper pillar with a bottom hexagonal shape of about 3 mm high, which can be attached to the screws and thus fixed to the trolley. The usefulness of these two copper pillars is to be able to place the fan module at a certain height from the floor of the trolley.

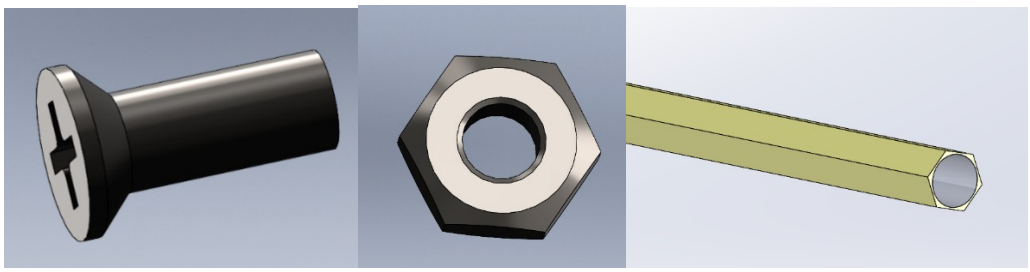


Figure 10 - nut and bolt

The next part is the PCB, which is exported directly from the PCB file, where there are two regulator parts that cannot be modified.

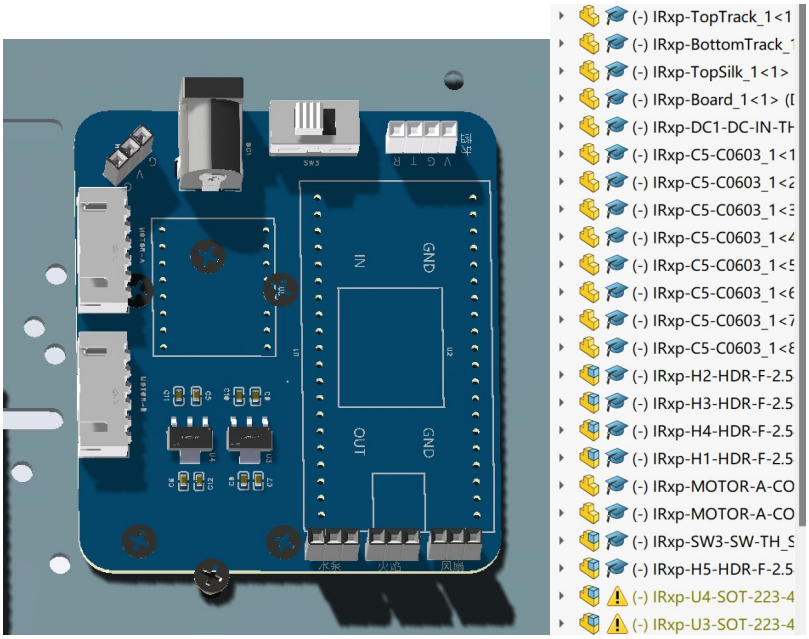



Figure 11 - PCB

The last part is the water pump, the type is the Submersible water pump, which is mainly composed of impeller, motor, mechanical seal shell and guide vane. Since the water pump is connected to the PCB board by wires, the water pump is related to the whole trolley mate.

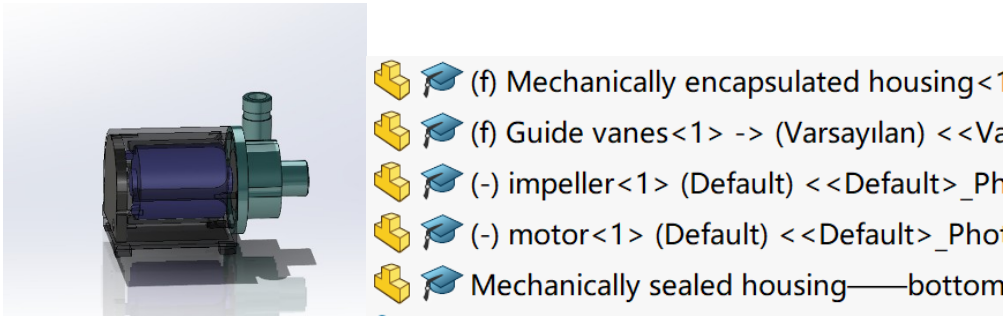



Figure 12 - Motor

The assembly of the trolley is to use various relationships to limit the position of each component and complete the mate according to the actual situation.

**Mechanical Model Structural Testing**

The entire structure operates with two distinctive sections; the lower is for the main chassis and wheels, and upper-housed various electronic components. In particular, the fan, flame sensor, PCB board, mbed microcontroller, and water pump are all secured onto the top of the base with the help of screws.

Of all these components, only the wheels and the fan can rotate. When the robot is in operation, the wheels should be able to rotate freely, not interfering with any other moving parts of the assembly. Thus, one of the major reasons while considering the design was that these blade tips would not strike any component at all.

Use the Motion study function in SolidWorks to simulate the movement process of the car. The specific configuration is as shown in the figure. A motor is added inside the fan to make it rotate at a speed of 100 rpm, which is roughly consistent with the actual speed.

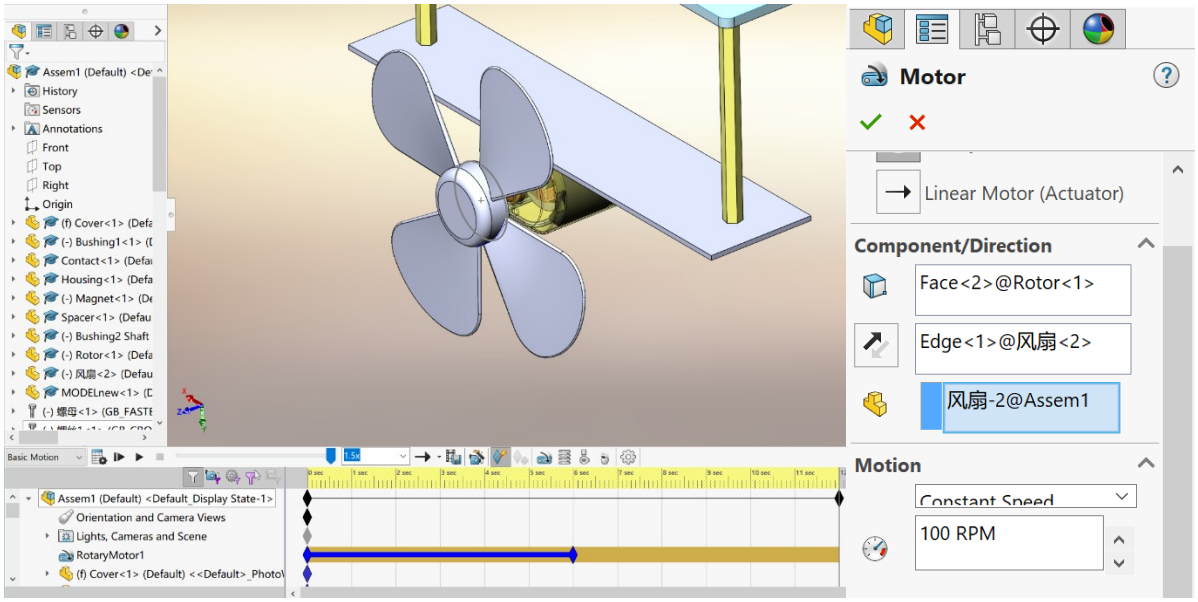



Figure 13 - Motion Study. a) fan analysis. b) configuration.

With the aid of the two long stick-like metal rods, the fan can be placed considerably high above the other inside device, guaranteeing the blades are sufficiently removed from any obstruction. One would consider it impossible for collision to occur, therefore affording this design to work as preferred.

When in practice, the fan blows out a gust quite strong enough to blow small flames out. The other dimension of suppression from the incorporated water pump also stops further spreading of the flames. The combined structure would then make this robot a substantially working and compact fire extinguishing system, more useful for demonstrations around small-scale emergency situations. The construction is sound, and all the moving parts are far enough apart to allow for their proper function, giving the overall assembly simplicity and rigidity.

**Mechanical system analysis**

During the winnable analysis of the mechanical structure of the robot, the authors largely focused on the two relatively rarely confronted normal work conditions, namely crash resistance and load-bearing

ability. Due consideration was given to two particular components: the iron plate, the top surface at the base, likely to be subjected to a collision, and the wheels and the connecting areas towards the motors, acting almost completely in load-bearing support. Cognizance of these critical areas helped funnelling much important contact structural testing without venturing into excessive complexity and additional burdens.

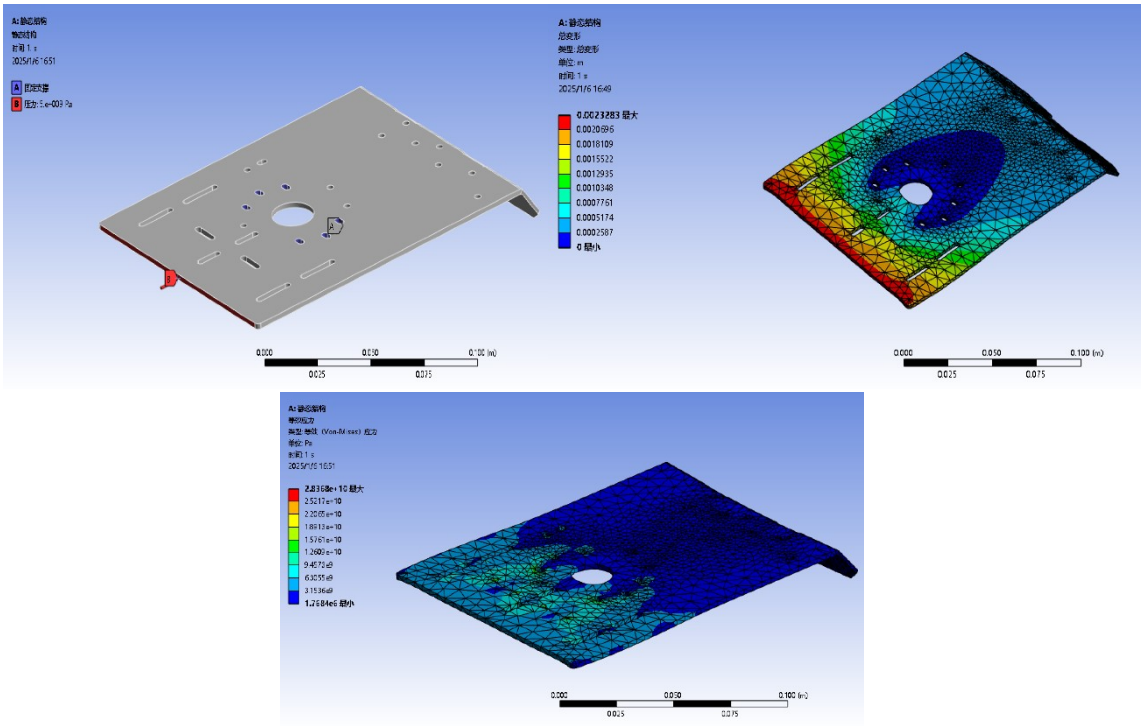


Figure 14 - Ansys analysis. a) Load and support. b) Deformation. c) von-mises stress

The modelling began with a high-fidelity three-dimensional model in SolidWorks that was imported into ANSYS Workbench. Having given the material properties of steel, representative of the actual construction of a robot, suitable boundary conditions such as fixed supports and pressure loads to simulate realistic collision and overload forces were applied. ANSYS Workbench produced deformation and stress contour plots for the iron plate in addition to that of the wheel-motor attachments. With great external forces, these simulations suggested both components underwent little deformation, making them indeed suitable for impact and heavy-load resistance under ordinary working conditions.

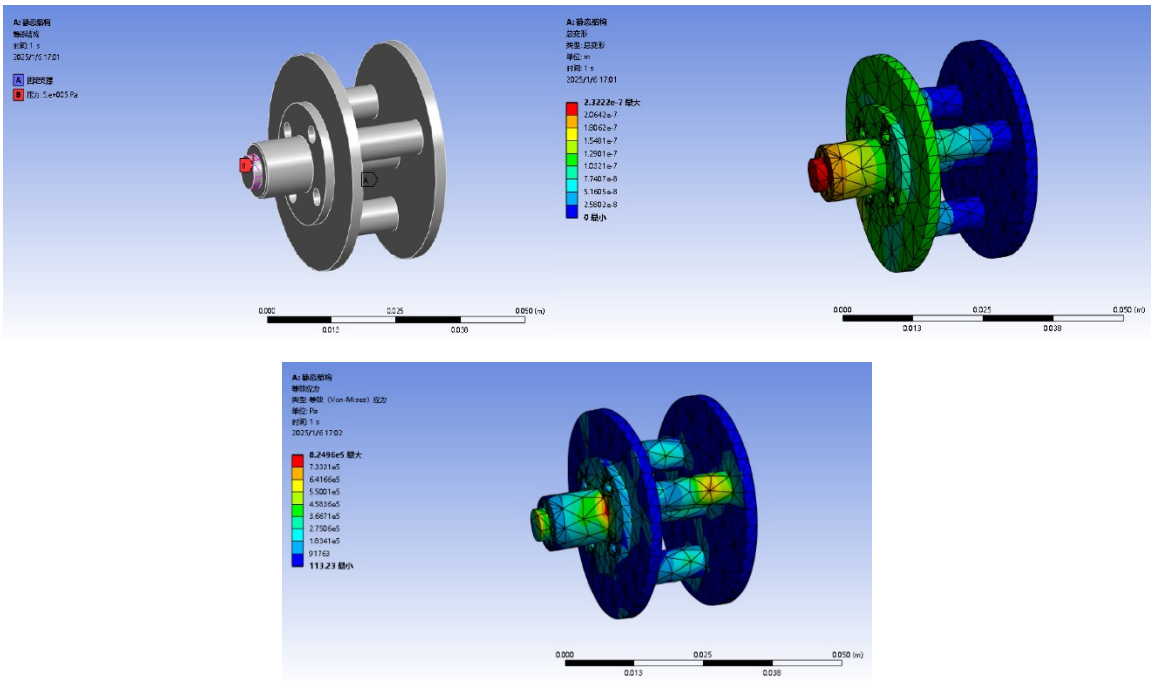


Figure 15 - Ansys analysis. a) Load and support. b) Deformation. c) von-mises stress.

However, while these results surely indicate that the materials and design selected are mechanically solid, the veracity of the robot's operation is beyond pure material characteristics. One must consider, for example, whether the motors can provide sufficient torque or how the life of the battery would be affected by the increased weight of the load. Hence, to prove the performance in the real world and corroborate the results of the simulation, a further test should be conducted on the overall system of the robot, such as the powertrain and the control. This dual approach would sign for the final design of adequate structural rigidity and functional competence in the light of varied conditions it is likely to meet.

### Microcontroller Implementation

This part is talking about the combination of code and circuit, the code is just like the brain of the main board, controlling the information transmission and data processing between various components on the circuit board, determining different modules how to react to the digital signals.

Firstly, it is needed to define the digital input and output ports, so that the main board can receive digital signals from other modules and send digital instructions to them.

```
// 定义引脚
DigitalIn FireSensor(p5); // 火焰传感器引脚
DigitalOut relayPump(p6); // 水泵继电器控制引脚
DigitalOut relayFan(p7);  // 风扇继电器控制引脚

MotorController LetfMotor(p21, p23, p22);  // 左电机控制对象
MotorController RightMotor(p26, p24, p25); // 右电机控制对象
BufferedSerial bluetooth(p9, p10, 9600);   // HC-05 蓝牙模块引脚 (TX, RX)
```

Figure 16 - Definition of input and output ports

About the fire sensor, it utilizes the thermal radiation of flames, when a flame is detected, it will send a low-level signal to the input port p5, otherwise it will send a high-level signal. Since the fan and water pump are triggered by low-level signal, when p5 port receives a low-level signal from the fire sensor, the main board will send a low-level signal pump from p6 and p7 ports which are connected to the fan and water pump, therefore activating them when detecting a flame.

The left motor is controlled by the p21, p23 and p22 output ports, they are all firstly connected to the motor drive module. Among which p21 is connected the PWM A port of the motor dive module, it outputs the PWM signal like frequency and duty cycle to determine the speed of motor. P23 and p22 ports are respectively connected to AIN1 and AIN2 ports of the motor drive module. The motor drive module receives low power digital signals from the main board and converts them to into current and voltage signal suitable for motor drive. The output ports on the motor drive module AO1 and AO2 are directly connected to the left motor. When the left motor is demanded to rotate forward, p23 port outputs a high-level signal, instead, p22 outputs a low-level signal, after the transformation occurred in the motor drive module, it will generate a voltage signal and the two output ports to control the motor to rotate forward. When it is demanded to rotate reverse, p23 port will output a low-level signal and p22 port will output a high-level signal. As for the right motor, it is controlled by output ports p26, p24 and p25 ports, among which p26 outputs PWM signal and p24, p25 ports control the rotation direction.

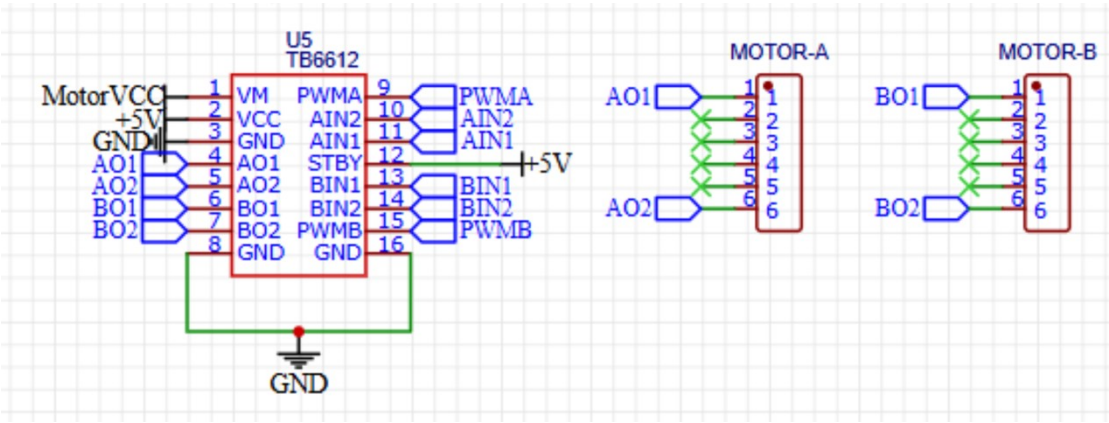



Figure 17 - Motor drive

The Bluetooth module's transmitter and receiver ports are relatively connected to p9 and p10 and the baud rate is set to 9600. P9 corresponds to the receiver port and p10 corresponds to the transmitter port, they are defined in the code with the use of 'BufferedSerial' class, which is introduced in Mbed os 6 for serial communication. By applying the 'read' function to receive the instruction from controller through Bluetooth. The car's movement is controlled using a smart phone, when using the smart phone to send an instruction to the car through Bluetooth, a character string will be sent. Different instructions like moving forward, turning left send different character strings, and by distinguishing different first character of different strings, Bluetooth module can tell the what the instruction is and Convert it into a digital signal and transmit it back to the main board, therefore drive the motors to operate to realize the expected movement.

**Electrical Circuit and PCB Design**

## 1. Main Port schematic Introduction

### a) Interaction with mbed Master Board

As the figure shows, this is the overall schematic. Mbed Master board was connected to ground and Vin at first. Then, the purpose was to use flame heat as input data to determine whether to extinguish the fire. Therefore, the P5 pin was set as SPI MOSI in the main mode to receive data from the temperature sensor. The P6 and P7 pins were then used as output pins in the main mode to send signals to the water pump and fan. The Bluetooth input and output interfaces were connected to the P9 and P10 pins, serving as the Serial port. This allowed the HC05 Bluetooth module to establish a connection between the main control board and a mobile phone for communication. Additionally, the two motors were wired to specific pins on the mbed PWMOUT pins in sequence. By controlling the frequency, the motors were able to perform various functions such as steering, moving forward, and stopping.

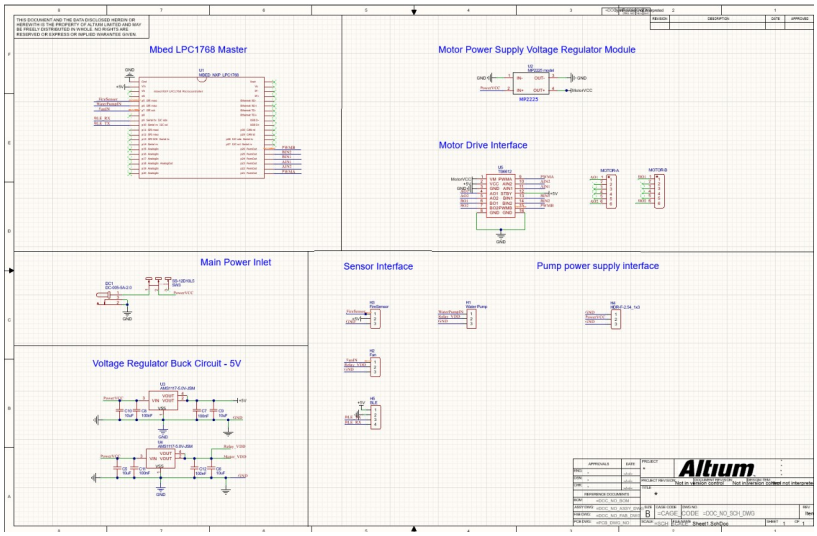



Figure 18 - Schematic

### b) Motor Power Supply Voltage Regulator Module

As the Power VCC may float up and down causes the motor output power to float, DC Regulator Power Supply Buck Module is purchased. As searched on the internet for relevant information, TB6612 only work when PIN1 VM of TB6612 is connected to the range of voltage 4.5V and 15V. Thus, this module was set to bring the voltage drop from unstable Power VCC to stable Motor drive voltage 7.4V.

### c) Motor Drive Interface Module

In this module，from P4 to P7 as signal output port were connected to P1 and P6 in Motor A and B. Then, Motor can be driven when AO1 and AO2 have voltage difference. while, from P9 to P11 and P13 to P15 were connected to mbed master board which is introduced in the Interaction with mbed Master Board part.

### d) Main Power Inlet Module

Main power inlet model concludes electric socket DC005 and single-pole, double-throw switch SS-12D10L5. The electric socket has three ports, P1 and P2 were connected to the circuit and P3 was set to ground.

### e) Voltage Regulator Buck Circuit Module

Voltage Regulator Buck Circuit Model AMS1117-5.0V-JSM was a Linear Voltage Regulators which can drop voltage from Power VCC to 5V. As, there are lots of element need 5V input, two Linear Voltage Regulators were used. One output 5V to master board, motor, BlE and temperature sensor. While, another one was used to output 5V to Fan and pump. Actually, during the commissioning phase found that the current of this module did not match the circuit which lead to some error. In addition, as shown in the manual this element also requires connecting 10uF capacitors for all application circuit to assure circuit stability.

### f) Sensor Interface and Pump Power Supply Interface Module

In this part, fire sensor, Fan and BLE were connected as the figure shows to let fire sensor accept user inputs. Then, pump or fans are interfaced with the device outputs. But after receiving the water pump it was unluckily found that the pumps we purchased could be driven directly from a DC power supply rather than from the mbed board. Based on this problem, after checking on the internet, it could be solved by connecting the pump to a relay first and then to the mbed main control board to achieve the goal. Thus, H1 was purchased as relay element which was connected to both of Water Pump Voltage IN and Relay VDD from the mbed board.

## 2. PCB Layout Design and Fabricated

The PCB component diagram of the purchased components is imported into the PCB library, and then according to the schematic diagram, the pins of each module are sequentially set to different nets such as GND, POWERVCC and so on. Components in accordance with the interface or with external components connected to the components such as mbed core board, motor drive module try to set the principle of the outside of the PCB board to be placed in order, and according to the tips of the white line for the connecting line laying copper. As the laying of lines should try to follow the principle of short distances and lines do not overlap with each other, part of the possible overlap of the line drawn on the bottom layer as figure shows.

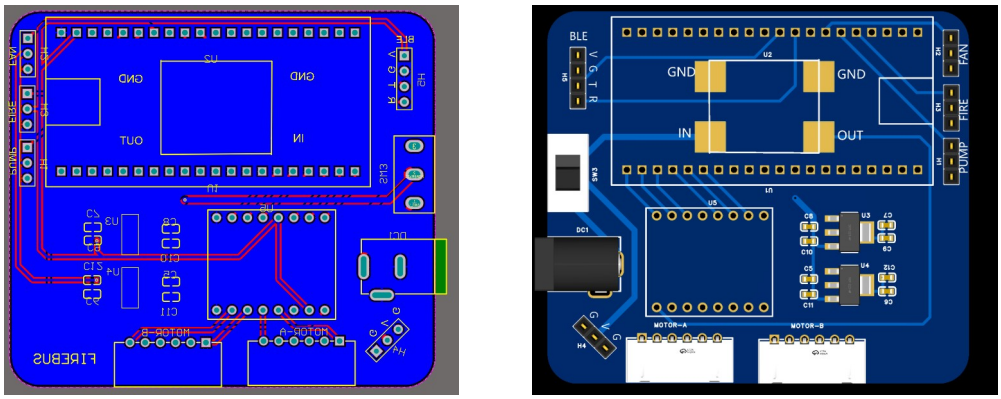



Figure 19 – a) back side PCB Board. b) 3D PCB

In addition, when repouring selected must pay attention to retain U2, U3 retain top solder and top paste, so that it does not cover the green oil leakage pads in the manufacture. After design rule check, no shorts or unconnected circuits were found, but some clearance issues were left. Thus, change the rule according to the requirement. When fabricating the PCB boards, due to sufficient time, chose a more high-performance black color board and completed soldering Voltage Regulator Buck Circuit Model and DC Regulator Power Supply Buck Module on the pads at the same time. Finally, PCB board was fabricated as figure shows.

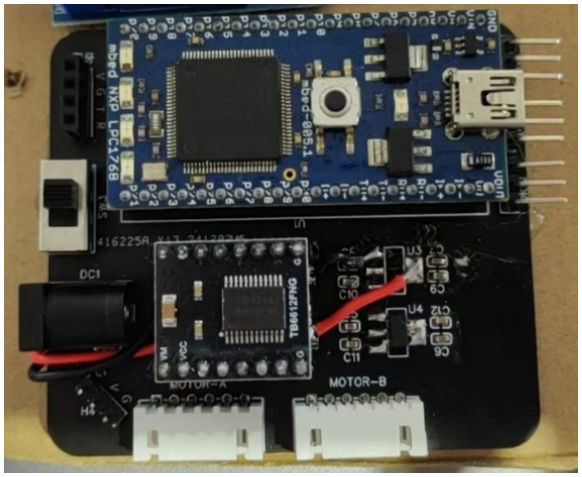



Figure 20 - Fabricated PCB

**Software Development**

The automatic fire car is intended to replace part of human firefighting work, lowering the risk of firefighting. The fire car's action like moving forward, backward and turning is controlled using Bluetooth, through which both trolly motors can be controlled. Based on existing conditions, mobile phone is applied. Besides, the function of automatic fire extinguishing is achieved by applying fire sensor, it utilizes the highly sensitive nature of infrared radiation to flame. When detecting the flame, the fire sensor immediately transmits back the high-level signal to activate the water pump and fan to put out the flame. According to above content, a flowchart about the working principle of the fire car can be drawn.

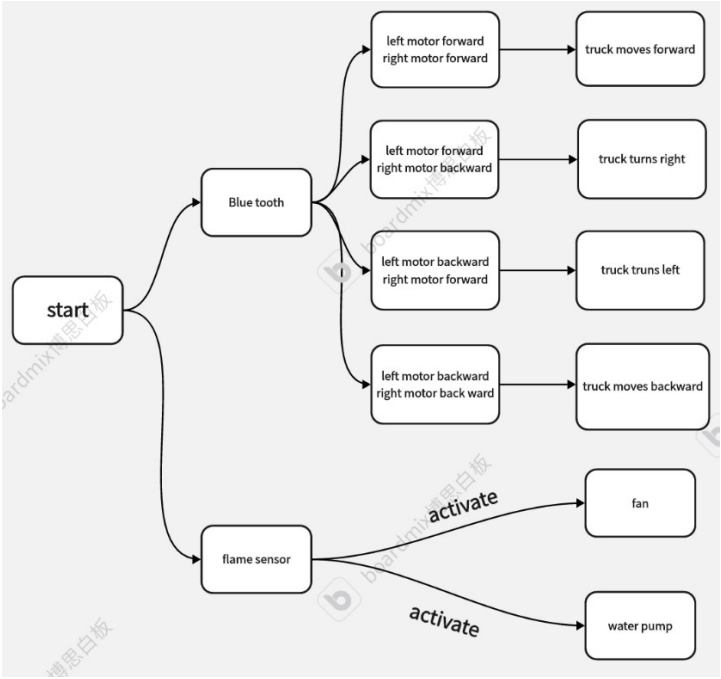



Figure 21 - Flowchart of fire car's working principle

The operation of a single motor, including moving forward, backward and staying still, is determined by the direction of current. Hence, by controlling two motors' operation, the car's movement like moving forward or backward, turning left or right can be achieved. However, before controlling both motors to realize these functions, firstly it is needed to define three states of a single motors, which are rotating forward, rotating reverse and staying still. In the function that controls the motor, use the 'enum' function to define these three states, in each of the state set the direction of the current to determine the direction that a single motor rotates or stay still. By combining different states of two motors, the car's movement like moving forward, backward and turning can be realized. Bluetooth module is used to control the motors' operation, use smart phone to send character strings to the Bluetooth module, by distinguishing different first characters of different strings, send different instructions to the motors to control their operations like moving forward and turning. With the use of "if…else" function can make this come true. For example, when receiving a string whose first character is 'k', both of the motors will rotate forward, therefore the car will move forward.

```
// 前进 字符 k
if (read_buff[0] == 'k')
{
    motorSpeed = 0.5;                                          // 设置电机速度为 0.5
    LetfMotor.setSpeed(SpeedDirection_Forward, motorSpeed);  // 设置左电机速度为 0.5
    RightMotor.setSpeed(SpeedDirection_Forward, motorSpeed); // 设置右电机速度为 0.5
}
```

Figure 22 - Bluetooth moving forward instruction

Now the problem of moving direction of motors has been solved, to control the speed of motors, PWM(pulse width modulation) is utilized. PWM is a method of digital encoding of analog signal levels, the basic principles of which is adjusting the average input voltage of the motor by changing the duty cycle of the PWM signal. The duty cycle means the ratio of high-level time in the pulse wave to the total period, which determines the power of the motors. The larger the duty cycle of the PWM signal, the higher the average input voltage value of the motor, therefore the faster the motor speed.

The frequency of the PWM signal determines the speed of signal change. High frequency PWM signals can control the level of analog signals more precisely.

through adjusting the duty cycle and frequency of the pulse signal, to realize the precise control of motor speed. Through testing, the frequency is finally set to 20kHz by setting the period to be 50us, and the duty cycle is set to 50%.

```
m_pwm.period_us(50);  // 20kHz = 1/50us
```

Figure 23 - Frequency of PWM

```
// 定义控制参数
float motorSpeed = 0.5; // 电机速度 (0.0 - 1.0)
```

Figure 24 - Duty rate of PWM

To make the Bluetooth control closer to people's usage habits, that when pressing a command button like 'moving forward', the car will react immediately. If no Bluetooth data is read, the car will stop and stay still. To realize this function, a variable 'count' is introduced. Define the initial value of variable 'count' to be zero, when no Bluetooth data is recieved, the value of 'count' will increase by one every 10ms, once the value of 'count' increases to 50, both motors will immediately stop their action. Besides, once Bluetooth data is read, the 'count' will be reset to zero, which means that the variable 'count' again increases from its initial state zero every time a instruction is received. The introduction of variable 'count' makes the operation of car much easier.

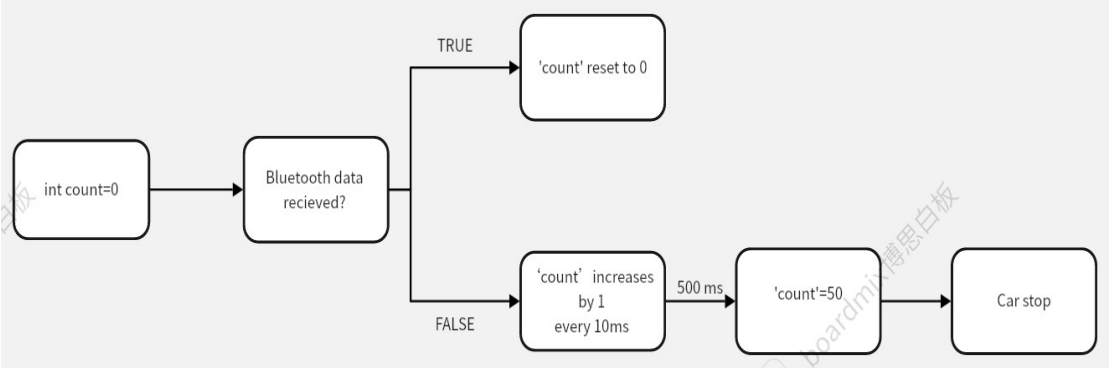



Figure 25 - Flowchart of variable 'count'

The flame detection and fan, water pump's activation are realized using the flame sensor, the sensor can detect the thermal radiation of the flame. When detecting the flame, the sensor will output a low-level signal, otherwise high-level signal. Hence, utilizing the 'if else' function, when receiving a low-level signal, activate the fan and water pump to put out the flame.

To improve the coding efficiency and concise, the function that controls the motors is independently written in a hpp file, "MotorController.hpp". This file contains functions that control an independent motor, to determine whether it rotates forward, reverse, or stop. And it also defines the frequency and duty cycle of PWM that controls the speed of motors Therefore, using the "#include

'MotorController.hpp'" instruction at the beginning of the main file to call the functions that control the motors, which makes the code much more concise and lowers the time cost to debug.

The codes of *main.cpp* and *MotorController.hpp* are displayed in the appendix part.

**Experiments and Results**

Since this is a multi-functional fire extinguishing trolley, the whole system consists of several subsystems. Therefore, it is necessary to test the function of a single subsystem first, and then carry out a combined test of the subsystem.

**1. Motor testing**

Because it is necessary to realize the function of a single-chip microcomputer, control the motor through PWM, and then control the speed. Therefore, this part needs to test whether the speed of the motor is properly controlled and whether it meets expectations.

By adjusting the space ratio of the PWM signal, the average voltage at both ends of the DC motor can be changed to realize the speed regulation of the DC motor. The motor model is MG513, DC geared motor, and the speed before deceleration is 11000rpm, and the speed after deceleration is 366±26rpm, and the frequency of the output PWM signal is 20kHz, and the air ratio of the output PWM signal is 0.5. The Figure 1 below is the specific model of the motor.




Figure 26 - Specific model of the motor

The test results show that the single-chip microcomputer can drive the motor to realize the movement of the trolley.

**2. Bluetooth module testing**

This part is to realize the function of Bluetooth module and APP communication, that is, to design the communication protocol and analyse the communication protocol to obtain the target data. The content of the test is whether the microcontroller correctly receives the control command sent by the APP. The following Figure 2 is the control interface of the Bluetooth software, through the test, the car can receive the instructions of the software to achieve movement




Figure 27 - The Bluetooth APP control interface

### 3. Flame sensor testing

This part is to achieve the status acquisition function of the flame sensor, that is, to test whether the microcontroller can correctly receive the high-level and low-level data returned by the sensor in the presence of flames. In the test, when the flame is close to the flame sensor, a red LED light lights up, indicating that the signal has been received, as shown in the Figure 3.

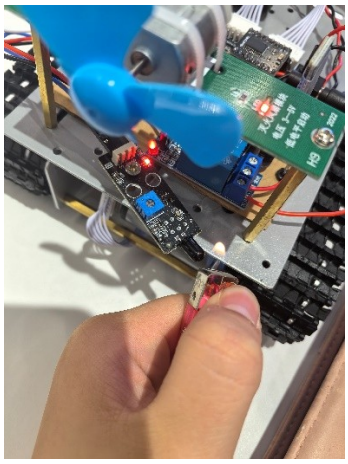


Figure 28 - When the flame close to the flame sensor

### 4. Fan testing

This part is to realize the control function of the fan, and test whether the start-stop state of the fan can be directly controlled in the program, to realize the fire extinguishing function. The interface that controls the fan can receive the signal from the flame sensor and output the signal to start and stop the fan, but in this test, only the subsystem function needs to be tested, so only the signal to control the start and stop of the fan needs to be artificially output. As can be seen from Figure 4, the test result is that the motor of the fan part can be directly controlled, and the rotation of the fan blades can be controlled.

However, during the subsequent test, the two blades of the fan broke shown in Figure 5, resulting in a decrease in the fan wind speed, which may affect the subsequent fire extinguishing test results.

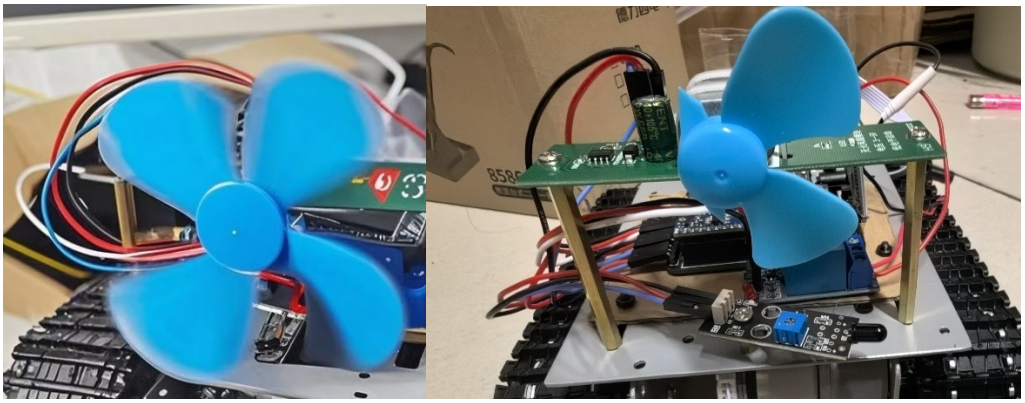


Figure 29 - The turning fan and Dropped two blades of the fan

## 5. Water pump testing

This part realizes the control function of the water pump, the water pump is controlled by the relay, and the start and stop state of the water pump can be directly controlled inside the program. Like the fan, it is also the fire extinguishing function of the trolley, and the interface that controls it can receive signals and output signals. In this test shown in Figure 6, the pump was able to receive the output signal and work normally.




Figure 30 - The water pump working properly

## 6. Integration testing

After testing that all subsystems can run normally and independently, the functions of all subsystems are combined to realize the overall control logic, and the overall functions are tested to see if the expected effect is met.

This fire extinguishing trolley mainly has the following two functions that need to be realized.

One is the trolley movement controlled by the motor drive and the Bluetooth module. This part is mainly to give instructions by Bluetooth, control the motor to rotate in reverse, and realize the forward and backward of the trolley and turn left and right. The motor is added to the left front wheel and the right rear wheel of the trolley, and they drive the wheel to rotate so as to realize the rotation of the track. When the left front motor and the right rear motor are rotating forward at the same time and the speed is the same, the car moves forward. When the left front motor and the right rear motor are reversed at the same time and the speed is the same, the trolley retreats. When the left front motor and the right rear motor brake at the same time, the trolley stops. When the left front motor is turning and the right rear motor is reversed, the trolley turns right. When the left front motor reverses and the right rear motor rotates forward, the trolley retreats. In this test, the car was able to achieve normal movement

The second is the fire extinguishing function, which is implemented by a flame sensor, a fan and a water pump. The flame is close to the flame sensor, the flame sensor receives the signal, backs it back to the single-chip microcomputer, and then outputs the signal to control the start of the fan and the water pump, so that the fan rotates and the water pump pumps. In this test, the fire extinguishing function of the car is sufficient

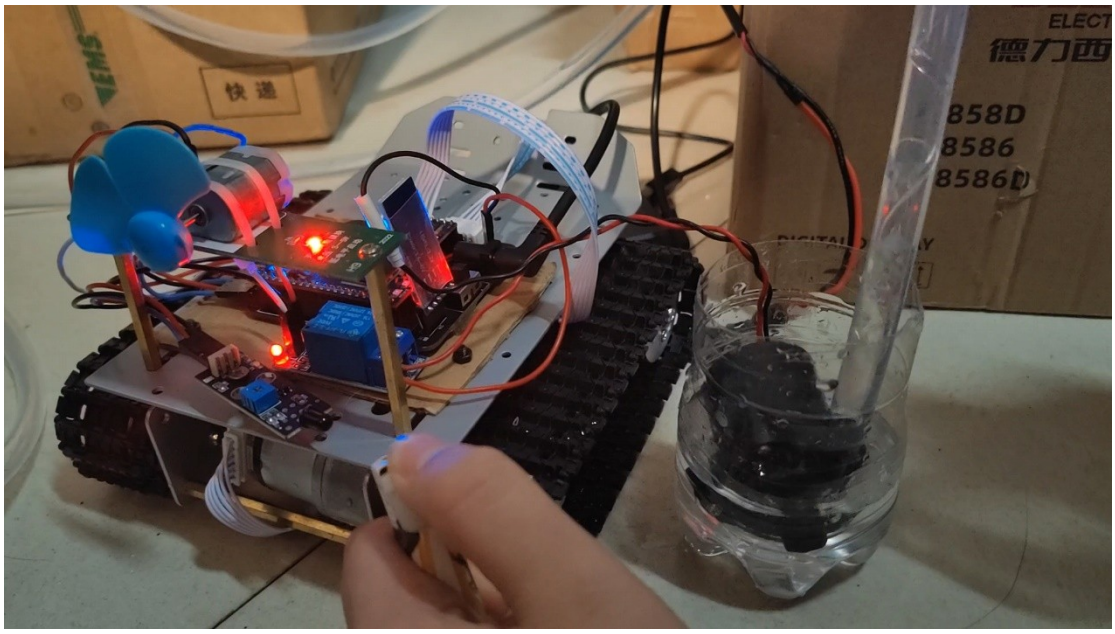



Figure 31 - Fire Testing

**Lessons Learned**

In the actual operation encountered a lot of problems, there are the following lessons and improvement plans

**1. Assembly problem**

In the process of assembling the model, I was not familiar with the structure splicing, and I found that some parts could not be installed after many times. In the next experiment, I

will use SolidWorks to decompose the model first and get familiar with the internal results of the model and the location of parts. It can also be spliced in order from the top down or from the bottom up, so that the installation is not inconvenient because the individual parts are in the middle

**2. Circuit problem**

When designing the power supply module, the maximum current required for the overall function is not considered, and the voltage regulator chip used can not meet the current supply, and the chip is hot. The problem was solved after replacing the high current regulator chip. In the next experiment, the maximum current and voltage of the circuit, as well as the maximum current of the constant voltage required by each component, will be carefully considered.

**3. Test problem**

In the subsequent integration test, the fan subsystem was accidentally damaged, and the operation could not be run. The motor was separated from the control board, resulting in the failure of the motor and the fan could not rotate, and the motor needed to be re-welded. In the next experiment, the use of wires to connect the motor and the control board will be given priority, because the motor is larger than the control board and easy to disconnect.

**Future Work**

Several avenues exist for upgrading and enhancing the functionality of the firefighting robot. One potential enhancement is the incorporation of an AI visual recognition module. This module would enable the robot to detect distant flames and potential fire hazards, thereby expanding its applicability to real-life fire scenes where the current flame detection system, which relies solely on proximity to the fire sensor, is limited.

Another area for improvement involves refining the robot's hardware components. Currently, the water pump is controlled via a relay element due to incompatibility with the microcontroller. Future iterations should aim to integrate a water pump that can be directly controlled by the mbed board, simplifying the system and reducing complexity. Additionally, attention should be paid to the relationship between the operating voltage or current of components and the circuit voltage or current to avoid errors encountered during the commissioning phase, such as the mismatch with the AMS1117-5.0V-JSM voltage regulator.

Cost reduction is also a consideration for future designs. Preliminary testing has shown that the use of black PCB versus ordinary green PCB does not significantly impact the project's performance. Therefore, adopting cheaper green PCB could be a viable option to lower overall costs. Furthermore, efforts should be made to streamline the component layout on the PCB to reduce wiring confusion and improve observability of interfaces, despite the challenges posed by the large number of electronic components.

The schematic design of the robot can be expanded to include additional features. Currently, it only fulfills the basic functions of flame recognition and fire extinguishment. Future work could involve integrating functionalities such as water level display and intelligent tracing to enhance the robot's capabilities and make it more versatile in various firefighting scenarios.

## Conclusion

Our team's project has developed a fire extinguishing car that can be linked with mobile phones. The user interacts with the mobile app to send electromagnetic signals captured by the vehicle's integrated Bluetooth module. This data is then transmitted to the mbed platform through the circuit and PCB board. Once powered on, the electronic components on the circuit board work together, with the processor processing data, memory storing information, and the input-output interface enabling external interaction. The mbed processes incoming data and executes commands based on programmed logic, allowing for motor control and fire extinguishing. During testing, we achieved remote control of the car to the ignition point. When the fan detects a fire source, it will automatically start the fire extinguishing work.

**References**

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[4] SHENPENG (2023) *Working principle of small water pump - Introduction to working principle and application scope.* Available at: https://www.spminipump.com/news/424.html#:~:text=The%20working%20principle%20of%20a%20small%20water%20pump,the%20pump%20body%20and%20then%20discharges%20it%20t. (Accessed: 3 January 2025)

## Appendix

### Appendix 1

Main.cpp

```
#include "mbed.h"
#include "MotorController.hpp"
#include <cstdint>
#include <cstring>
// 定义引脚
DigitalIn FireSensor(p5); // 火焰传感器引脚
DigitalOut relayPump(p6); // 水泵继电器控制引脚
DigitalOut relayFan(p7);  // 风扇继电器控制引脚

MotorController LetfMotor(p21, p23, p22);  // 左电机控制对象
MotorController RightMotor(p26, p24, p25); // 右电机控制对象
BufferedSerial bluetooth(p9, p10, 9600);   // HC-05 蓝牙模块引脚 (TX, RX)

// 定义控制参数
float motorSpeed = 0.5; // 电机速度 (0.0 - 1.0)

int main()
{
    relayFan = 0;  // 初始化风扇继电器为关闭
    relayPump = 0; // 初始化水泵继电器为关闭
    uint8_t read_buff[128];
    uint32_t count = 0;
    while (true)
    {
        if (bluetooth.readable())//判断蓝牙串口数据是否可读
        {
            memset(read_buff, 0, sizeof(read_buff));
            int len = bluetooth.read(read_buff, sizeof(read_buff)); // 读取蓝牙命令
```

```
    printf("len: %d, read_buff: %s\n", len, read_buff);
    if (len > 0)
    {
      count = 0;//读到蓝牙数据，计数器清零
      // 前进 字符 k
      if (read_buff[0] == 'k')
      {
        motorSpeed = 0.5;                                  // 设置电机速度为 0.5
        LetfMotor.setSpeed(SpeedDirection_Forward, motorSpeed);  // 设置左电机速度为 0.5
        RightMotor.setSpeed(SpeedDirection_Forward, motorSpeed); // 设置右电机速度为 0.5
      }
      // 后退 字符 m
      if (read_buff[0] == 'm')
      {
        motorSpeed = 0.5;                                  // 设置电机速度为 0.5
        LetfMotor.setSpeed(SpeedDirection_Backward, motorSpeed);  // 设置左电机速度为 -0.5
        RightMotor.setSpeed(SpeedDirection_Backward, motorSpeed); // 设置右电机速度为 -0.5
      }
      // 左转 字符 p
      if (read_buff[0] == 'p')
      {
        LetfMotor.setSpeed(SpeedDirection_Backward, motorSpeed); // 设置左电机速度为 -0.5
        RightMotor.setSpeed(SpeedDirection_Forward, motorSpeed); // 设置右电机速度为 0.5
      }
      // 右转 字符 q
      if (read_buff[0] == 'q')
      {
        LetfMotor.setSpeed(SpeedDirection_Forward, motorSpeed);   // 设置左电机速度为 0.5
        RightMotor.setSpeed(SpeedDirection_Backward, motorSpeed); // 设置右电机速度为 -0.5
      }
    }
  }
```

```
    else
    {
        count++;//计数器累加  10ms 累加一次
        if (count % 50 == 0)//累加到 50 次,也就是 500ms，判定为松开按键，控制停止
        {
            count = 0;
            LetfMotor.setSpeed(SpeedDirection_Stop, 0);  // 停止左电机
            RightMotor.setSpeed(SpeedDirection_Stop, 0); // 停止右电机
        }
    }

    // 检测火焰
    if (FireSensor == 0)
    {               // 火焰传感器检测到火焰
        relayFan = 0;  // 打开风扇
        relayPump = 0; // 打开水泵
    }
    else
    {
        relayFan = 1;  // 关闭风扇
        relayPump = 1; // 关闭水泵
    }

    ThisThread::sleep_for(1ms); // 等待一段时间  1ms 实际是 10ms
  }
}
```

## Appendix 2

MotorController.hpp

```
#ifndef MOTORCONTROLLER_HPP
enum SpeedDirection
{
    SpeedDirection_Forward = 0,
    SpeedDirection_Backward = 1,
    SpeedDirection_Stop = 2
};
class MotorController
{
private:
    /* data */
public:
    MotorController(PinName pwm, PinName left, PinName right):
        m_pwm(pwm),m_left(left),m_right(right)
    {
        m_pwm.period_us(50);  // 20kHz = 1/50us
    }
    void setSpeed(SpeedDirection speed_dir,float speed)
    {
        if(speed_dir == SpeedDirection_Forward) //正转
        {
            m_left = 1;
            m_right = 0;
        }
        else if(speed_dir == SpeedDirection_Backward) //反转
        {
            m_left = 0;
            m_right = 1;
        }
```

```
    else //停止
    {
        m_left = 0;
        m_right = 0;
    }
    m_pwm.write(speed);
  }
  ~MotorController();
private:
  PwmOut m_pwm;
  DigitalOut m_left;
  DigitalOut m_right;
};


#endif // !MOTORCONTROLLER_HPP
```